LAB RECORD

Computer Hardware, Operating System
& Networking Practicals

Experiments 2 to 13

Each experiment contains: Aim, Requirements, Procedure, Diagram / Screenshot, Observation, Result and Viva Questions

Name:	Rajani	**Register No.:**	_____
Branch:	_____	**Semester:**	_____
Faculty:	_____	**Date:**	_____

Exp 2	Unmounting the PSU and identifying output connectors
Exp 3	Motherboard components and labeled layout
Exp 4	Power management using BIOS/UEFI
Exp 5	OS settings: screen orientation and microphone test
Exp 6	Encrypting and decrypting files/folders using CLI
Exp 7	Troubleshooting startup issues (WinRE tools)
Exp 8	Command-line disk management tools
Exp 9	Process and performance monitoring
Exp 10	Network configuration (IP, DNS, adapter)
Exp 11	Installing and using a VPN
Exp 12	E-waste management (case study)
Exp 13	Computer laboratory setup plan

EXPERIMENT 2

Unmounting the PSU and Identifying Output Connectors

AIM: To unmount the Power Supply Unit (PSU) from the PC cabinet and identify its different output connectors.

REQUIREMENTS / APPARATUS

Desktop PC cabinet, Phillips screwdriver, anti-static wrist strap, small container for screws, PSU.

PROCEDURE

1. Shut down the PC, switch off the SMPS rear switch and remove the power cable from the wall socket.
2. Wait 1-2 minutes, then press the power button once to discharge the residual charge.
3. Wear the anti-static wrist strap and place the cabinet on a flat table.
4. Remove the side panel (usually 2 screws at the back).
5. Disconnect the PSU cables one by one: 24-pin motherboard, CPU 4/8-pin, SATA power, Molex and PCIe (if present).
6. Unscrew the 4 screws holding the PSU at the back of the cabinet (keep one hand supporting the PSU).
7. Slide the PSU out gently and place it on the table.
8. Read the PSU label (input/output ratings) and identify each connector by shape and number of pins.

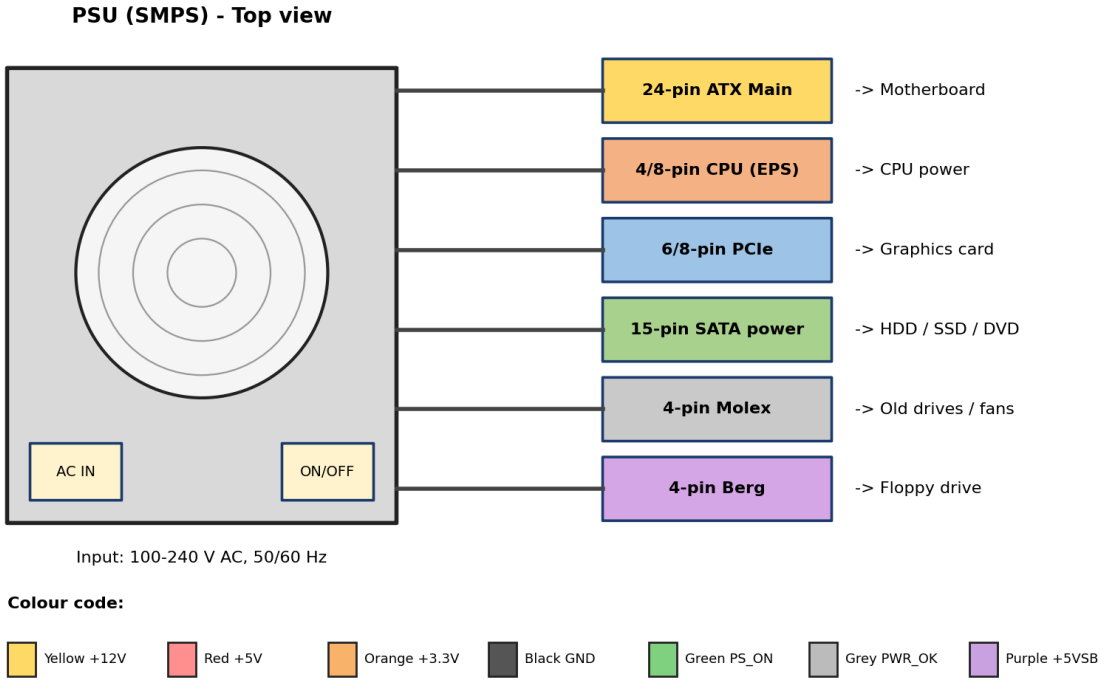


Fig 2.1: PSU output connectors and their uses

OBSERVATION

Connector	Pins	Connected To	Wire colours
ATX Main	24 (20+4)	Motherboard	Yellow, Red, Orange, Black, Green, Grey, Purple
CPU / EPS	4 or 8	CPU power on motherboard	Yellow, Black
SATA power	15	HDD / SSD / DVD drive	Yellow, Red, Orange, Black

Connector	Pins	Connected To	Wire colours
Molex (peripheral)	4	Old IDE drives, fans	Yellow, Red, Black
Berg (floppy)	4	Floppy drive	Yellow, Red, Black
PCIe	6 / 8	Graphics card	Yellow, Black

Wire colour	Voltage / Signal
Yellow	+12 V
Red	+5 V
Orange	+3.3 V
Black	Ground (0 V)
Green	PS_ON (short to ground to switch the PSU on)
Grey	Power Good (PWR_OK)
Purple	+5 V Standby

PSU label reading: Input 100-240 V AC, 50/60 Hz. Output power rating = _____ W (note it from your PSU).

Note: Never open the PSU casing. Capacitors inside can hold a dangerous charge even after unplugging.

RESULT: The PSU was unmounted safely and its output connectors were identified along with their voltages and uses.

VIVA QUESTIONS

Q1. **What does a PSU do?**
Ans: It converts AC mains supply into the low-voltage DC needed by the PC components.

Q2. **Full form of SMPS?**
Ans: Switched Mode Power Supply.

Q3. **Which connector powers the motherboard?**
Ans: The 24-pin ATX connector.

Q4. **What is the function of the green wire?**
Ans: PS_ON. Shorting it to ground turns the PSU on.

Q5. **Why is an anti-static strap worn?**
Ans: To prevent ESD (electrostatic discharge) damage to components.

EXPERIMENT 3

Motherboard Components and Labeled Layout

AIM: To identify the various components of the motherboard and draw a neat, labeled motherboard layout.

REQUIREMENTS / APPARATUS

Motherboard, anti-static mat, pencil, scale, record sheet.

PROCEDURE

1. Place the motherboard on an anti-static mat and wear the wrist strap.
2. Identify the major parts one by one, starting from the top-left corner.
3. Read the names printed on the board near each component (e.g. PCIE1, DIMM_A1, SATA1, BAT1).
4. Note the form factor (ATX / micro-ATX / mini-ITX) and the board model printed on it.
5. Draw the board as a rectangle in the record and mark every component with a labeled arrow (as in Fig 3.1).

Motherboard Layout (ATX form factor)

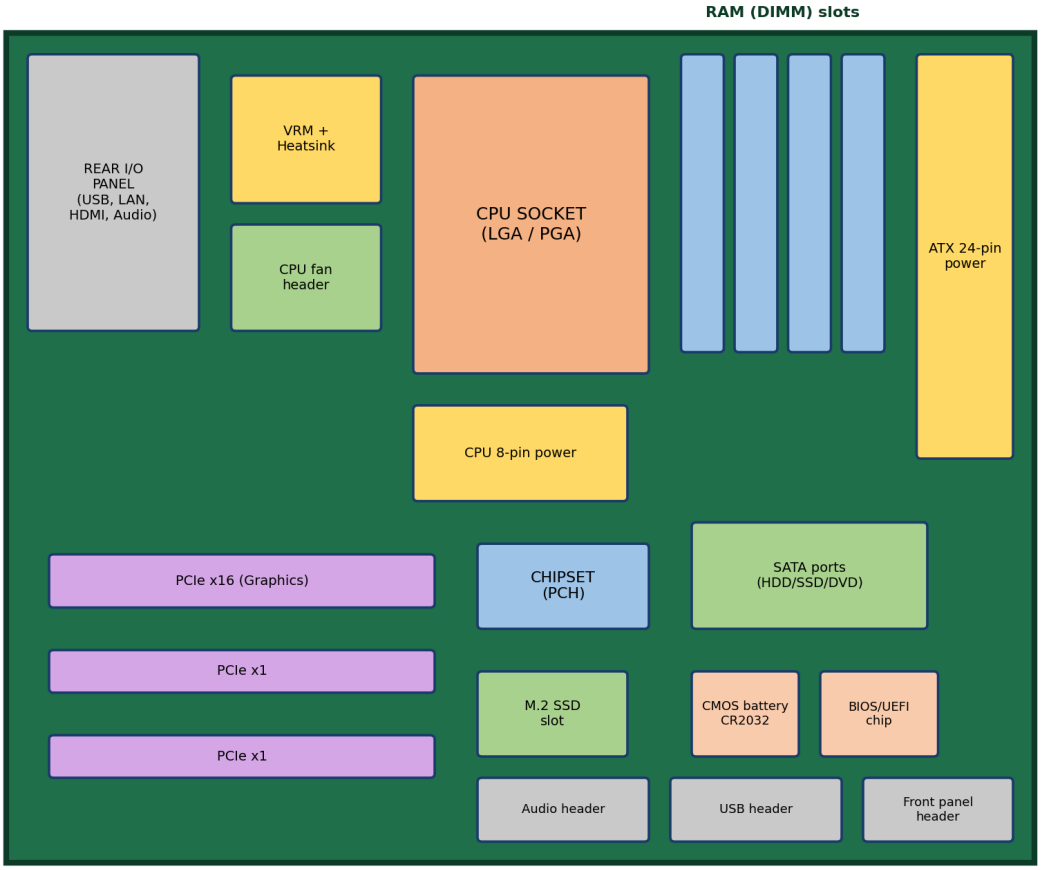


Fig 3.1: Labeled motherboard layout

OBSERVATION

Component	Function
CPU socket (LGA/PGA)	Holds the processor
RAM (DIMM) slots	Hold the memory modules
Chipset (PCH)	Controls data flow between CPU, RAM and peripherals

Component	Function
PCIe x16 / x1 slots	Graphics card, sound/LAN/other expansion cards
SATA ports	Connect HDD / SSD / DVD drives
M.2 slot	NVMe / SATA M.2 SSD
ATX 24-pin connector	Main power input from the PSU
CPU power connector (4/8-pin)	Dedicated power for the CPU
BIOS/UEFI chip	Stores the firmware
CMOS battery (CR2032)	Keeps BIOS settings and the clock
VRM + heatsink	Regulates voltage supplied to the CPU
Front panel header	Power/reset switch and LEDs
USB / audio headers	Front USB ports and audio jacks
CPU / system fan headers	Power for the cooling fans
Rear I/O panel	USB, HDMI/VGA, LAN and audio ports

Form factor: _____ Board model: _____ Socket type: _____

RESULT: The motherboard components were identified and a labeled layout was drawn.

VIVA QUESTIONS

Q1. **What is a motherboard?**
Ans: The main circuit board that connects all the components of a PC.

Q2. **Why is the CMOS battery needed?**
Ans: To retain BIOS settings and the time when power is off.

Q3. **Name some motherboard form factors.**
Ans: ATX, micro-ATX, mini-ITX.

Q4. **What is a chipset?**
Ans: A set of ICs that manages communication between the CPU, memory and I/O devices.

Q5. **Difference between DIMM and SO-DIMM?**
Ans: DIMM is used in desktops; SO-DIMM is smaller and used in laptops.

EXPERIMENT 4

Power Management Using BIOS/UEFI

AIM: To implement basic power management features using BIOS/UEFI settings.

REQUIREMENTS / APPARATUS

Desktop / laptop with BIOS/UEFI access, keyboard.

PROCEDURE

1. Restart the PC and press the BIOS key repeatedly (Del / F2 / F10 / F12 / Esc depending on the brand).
2. Use the arrow keys to open the **Advanced** / **Power** / **Power Management** menu.
3. Note the default value of each setting before changing it.
4. Set: ACPI Suspend Type = S3; Restore on AC Power Loss = Last State; Wake on LAN = Enabled; Resume by RTC Alarm = set a time; CPU C-States and Intel SpeedStep = Enabled; Fan control = Auto.
5. Press **F10** to save and exit.
6. After Windows starts, verify with `powercfg /a` (available sleep states) and, on a laptop, `powercfg /batteryreport`.

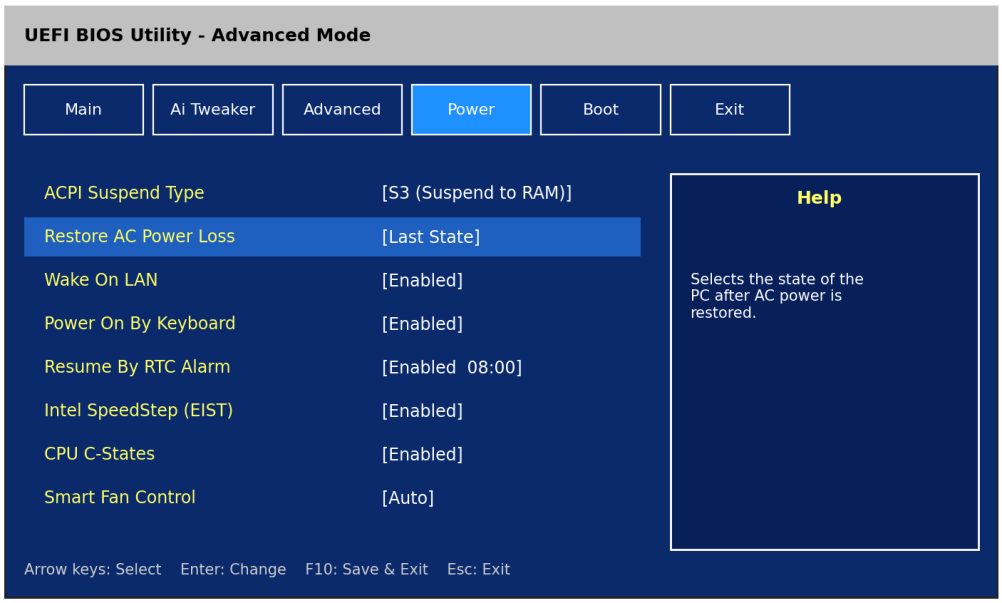


Fig 4.1: Power menu of a UEFI BIOS (sample screen; your screen may differ by brand)

OBSERVATION

Fill the Default column from your own BIOS screen.

Setting	Default	Changed to	Effect
ACPI Suspend Type		S3	Low-power sleep with RAM retained
Restore on AC Power Loss		Last State	PC returns to its previous state when power is restored
Wake on LAN		Enabled	PC can be woken by a network packet
Resume by RTC Alarm		08:00	PC starts automatically at the set time
C-States / SpeedStep		Enabled	CPU lowers speed and voltage when idle, saving power
Smart Fan		Auto	Fan speed follows temperature

RESULT: Basic power management features were configured successfully through BIOS/UEFI.

VIVA QUESTIONS

Q1. **What is BIOS?**
Ans: Basic Input Output System, firmware that initializes the hardware at boot.

Q2. **Difference between BIOS and UEFI?**
Ans: UEFI is the modern replacement with a GUI, support for GPT disks over 2 TB, Secure Boot and faster boot.

Q3. **What is ACPI?**
Ans: Advanced Configuration and Power Interface, the standard for OS-controlled power management.

Q4. **What is the S3 state?**
Ans: Sleep (Suspend to RAM): RAM stays powered, everything else is off.

Q5. **What is Wake on LAN?**
Ans: A feature that powers on the PC remotely using a special network packet.

EXPERIMENT 5

OS Settings: Screen Orientation and Microphone Test

AIM: To configure essential operating system settings: (a) inverting and rearranging the screen, (b) testing the microphone.

REQUIREMENTS / APPARATUS

Windows PC, monitor (second monitor optional), microphone / headset.

PROCEDURE (A): INVERTING AND REARRANGING THE SCREEN

1. Right-click on the desktop and select **Display settings** (Settings > System > Display).
2. Under **Display orientation** choose Landscape, Portrait, Landscape (flipped) or Portrait (flipped).
3. Click **Keep changes** within 15 seconds, otherwise the screen reverts automatically.
4. To rearrange multiple monitors, click **Identify**, then drag the monitor boxes into the required position. Under Multiple displays choose Extend or Duplicate.
5. Shortcut (Intel graphics): Ctrl + Alt + arrow keys. Press Ctrl + Alt + Up arrow to restore the normal orientation.

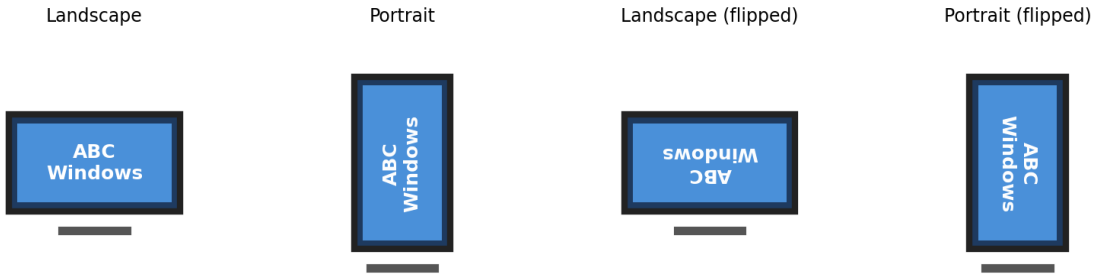


Fig 5.1: Display orientations available in Windows

PROCEDURE (B): TESTING THE MICROPHONE

1. Connect the microphone or headset to the correct jack / USB port.
2. Open **Settings > System > Sound > Input** and select the microphone under *Choose a device for speaking or recording*.
3. Speak and watch the **Test your microphone** bar move.
4. Alternative: open **Sound Recorder**, record a voice clip and play it back. (Control Panel > Sound > Recording > Properties > Listen also works.)
5. If there is no input, check Settings > Privacy > Microphone, the driver in Device Manager, and the cable.

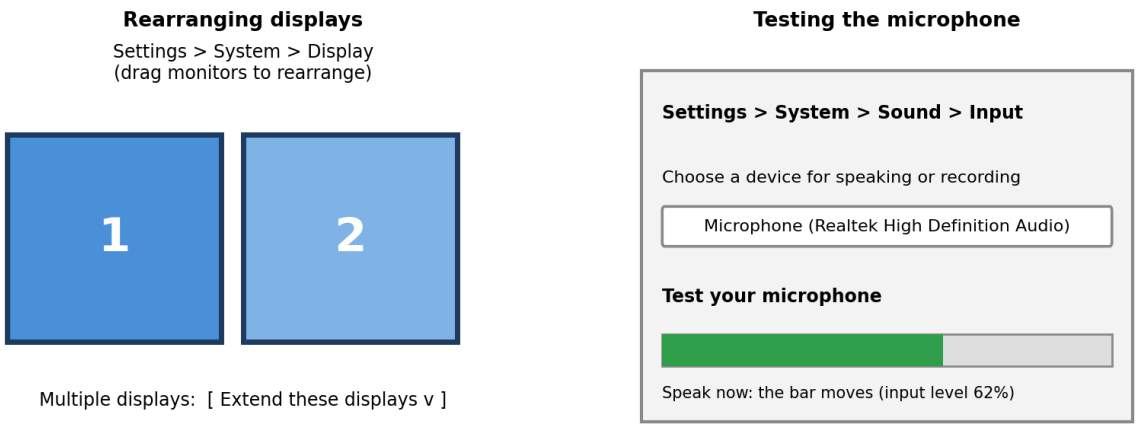


Fig 5.2: (left) rearranging displays, (right) microphone input test

OBSERVATION

Test	Observation
Landscape (flipped)	Screen rotated by 180 degrees; mouse movement also appears reversed
Portrait	Screen rotated by 90 degrees; suitable for reading documents / coding
Extend display	Desktop spread across two monitors; the order matches the box positions
Microphone level bar	Bar moves while speaking, so the microphone is working
Recording playback	Recorded voice clearly audible

RESULT: The screen orientation was changed and rearranged, and the microphone was tested successfully.

VIVA QUESTIONS

Q1. **Why can the screen be rotated?**
Ans: The graphics driver supports display rotation.

Q2. **What is the use of Portrait mode?**
Ans: Reading long documents and writing code.

Q3. **How do you update an audio driver?**
Ans: Device Manager > Sound, video and game controllers > right-click > Update driver.

Q4. **What is Extend display?**
Ans: The desktop is spread across multiple monitors as one large workspace.

Q5. **What will you do if the microphone does not work?**
Ans: Check the connection, default input device, privacy permission and driver.

EXPERIMENT 6

Encrypting and Decrypting Files / Folders Using CLI

AIM: To use the Command Line Interface (CLI) to encrypt and decrypt files and folders.

REQUIREMENTS / APPARATUS

Windows Pro/Enterprise/Education with an NTFS drive (for `cipher`), or a Linux terminal (for GPG / OpenSSL).

PROCEDURE A: WINDOWS (CIPHER COMMAND)

1. Create a folder `C:\Secret` and a file `notes.txt` inside it.
2. Open **Command Prompt**.
3. Encrypt a file: `cipher /e "C:\Secret\notes.txt"`
4. Encrypt a folder with its contents: `cipher /e /s:"C:\Secret"`
5. Check the status: `cipher "C:\Secret"` (E = encrypted, U = unencrypted).
6. Decrypt: `cipher /d /s:"C:\Secret"`
7. Optional key backup: `cipher /x backup_key`

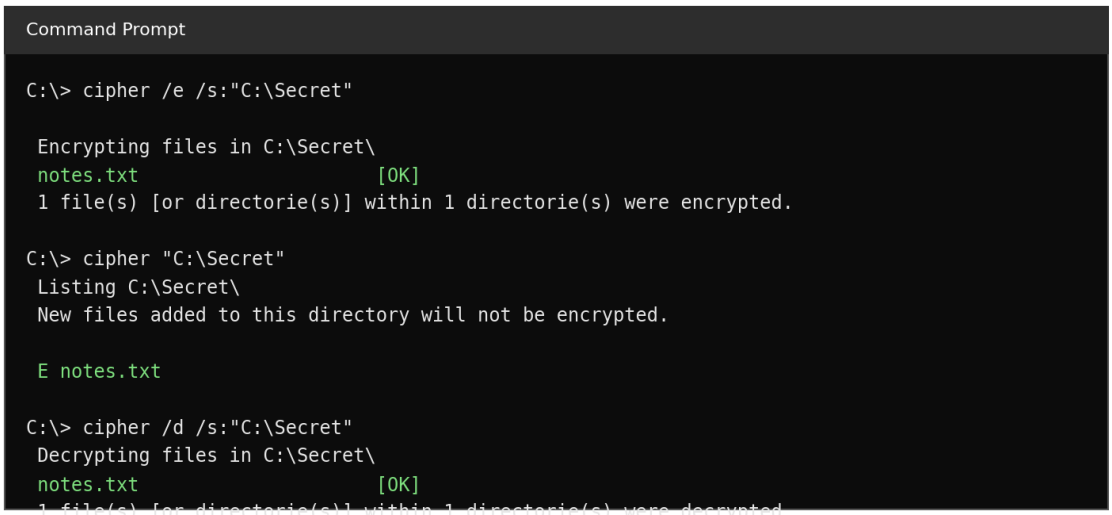


Fig 6.1: Encrypting, checking and decrypting a folder using cipher

PROCEDURE B: LINUX (GPG AND OPENSSL)

1. Create a file: `echo "hello" > file.txt`
2. Encrypt with GPG: `gpg -c file.txt` (enter a passphrase; *file.txt.gpg* is created).
3. Decrypt: `gpg -d file.txt.gpg > file_dec.txt`
4. OpenSSL option: `openssl enc -aes-256-cbc -salt -pbkdf2 -in file.txt -out file.enc`, then `openssl enc -d -aes-256-cbc -pbkdf2 -in file.enc -out file_out.txt`
5. For a folder, compress and then encrypt: `tar czf folder.tar.gz folder/ && gpg -c folder.tar.gz`

Linux Terminal

```
$ echo "hello" > file.txt
$ gpg -c file.txt
Enter passphrase: ********
$ ls
file.txt  file.txt.gpg
$ cat file.txt.gpg
?   ???~?*)??#?r(??    <- unreadable cipher text
$ gpg -d file.txt.gpg > file_dec.txt
gpg: AES256.CFB encrypted data
$ cat file_dec.txt
hello
```

Fig 6.2: File encryption and decryption using GPG in Linux

OBSERVATION

Step	Observation
After cipher /e	File name appears in green in Explorer; status shows E; another user account cannot open it
After cipher /d	Original content readable again; status shows U
After gpg -c	file.txt.gpg is unreadable binary data
After gpg -d	Original text 'hello' is restored

Note: EFS (cipher /e) does not work on Windows Home edition. Use BitLocker or the Linux method instead.

RESULT: Files and folders were encrypted and decrypted successfully using CLI commands.

VIVA QUESTIONS

Q1. **What is encryption?**
Ans: Converting plain text into unreadable cipher text using a key.

Q2. **What is EFS?**
Ans: Encrypting File System, a feature of NTFS that encrypts files per user.

Q3. **What is the use of the cipher command?**
Ans: It displays or alters the encryption of files and folders on NTFS.

Q4. **Symmetric vs asymmetric encryption?**
Ans: Symmetric uses one key (AES); asymmetric uses a public/private key pair (RSA).

Q5. **Why should the encryption key be backed up?**
Ans: If the key is lost the encrypted data cannot be recovered.

EXPERIMENT 7

Troubleshooting Startup Issues

AIM: To troubleshoot system startup issues using Safe Mode, Automatic Repair, System Restore, System Reset and System Image Recovery.

REQUIREMENTS / APPARATUS

Windows 10/11 PC or virtual machine (practise on a lab/VM only), Windows installation USB (optional), backup of important data.

Entering WinRE (Windows Recovery Environment): hold **Shift** and click Restart, or force-shutdown the PC 3 times during boot, or boot from Windows installation media and choose *Repair your computer*. Then go to **Troubleshoot > Advanced options**.

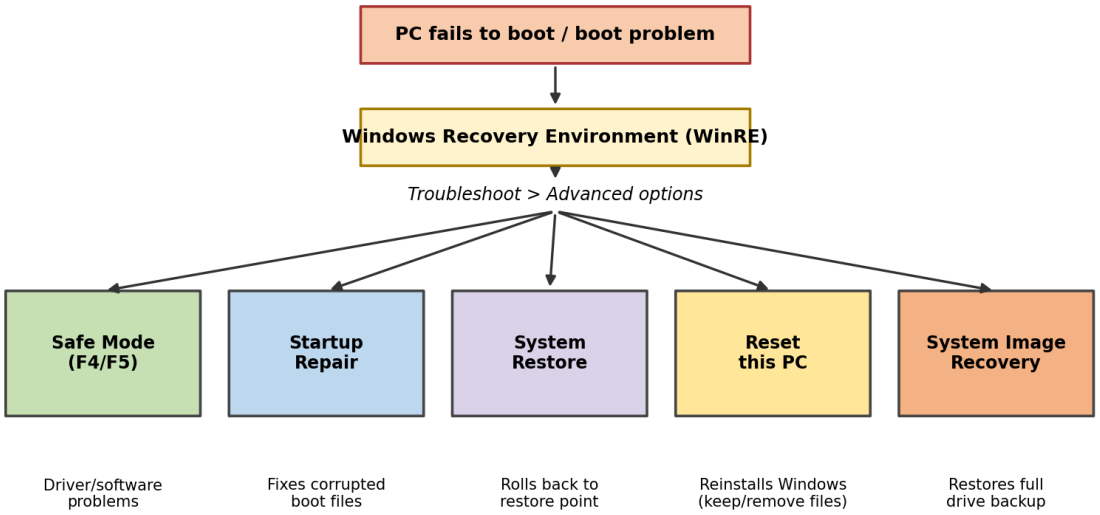


Fig 7.1: Recovery options available in the Windows Recovery Environment

PROCEDURE

1. **Safe Mode:** Troubleshoot > Advanced options > Startup Settings > Restart > press **F4** (Safe Mode) or **F5** (Safe Mode with Networking). Alternatively run `msconfig` > Boot tab > tick Safe boot. Only basic drivers load, so faulty drivers/software can be removed.
2. **Automatic (Startup) Repair:** Advanced options > Startup Repair. Windows diagnoses and fixes boot problems itself.
3. **System Restore:** Advanced options > System Restore (or run `rstrui.exe`). Choose a restore point, click Next > Finish. System files and settings roll back; personal files are not affected.
4. **System Reset:** Settings > System > Recovery > *Reset this PC*. Choose **Keep my files** or **Remove everything**, then Cloud download or Local reinstall.
5. **System Image Recovery:** Advanced options > System Image Recovery, select the backup image (created earlier using Backup and Restore (Windows 7)) and restore the entire drive.
6. **Manual boot repair and file checks** from the WinRE Command Prompt (Fig 7.2): `bootrec /fixmbr`, `/fixboot`, `/scanos`, `/rebuildbcd`, then `sfc /scannow` and `DISM /Online /Cleanup-Image /RestoreHealth`.



Fig 7.2: Boot repair commands in the WinRE Command Prompt

OBSERVATION

Method	When used	Effect on data
Safe Mode	Driver or software problem	No data loss
Startup Repair	Boot files corrupted	No data loss
System Restore	After a bad update / installation	Apps installed after the restore point are removed; personal files safe
Reset this PC	Severe OS corruption	Depends on the option (keep or remove files)
System Image Recovery	Complete system failure	Drive restored to image state; newer data lost

RESULT: Various startup problems were troubleshot using the recovery tools of Windows.

VIVA QUESTIONS

Q1. **What is Safe Mode?**
Ans: A diagnostic mode that loads only the minimum drivers and services.

Q2. **Difference between System Restore and System Reset?**
Ans: Restore returns the system to an earlier state; Reset reinstalls Windows.

Q3. **What is a restore point?**
Ans: A snapshot of system files and the registry at a particular time.

Q4. **What is MBR?**
Ans: Master Boot Record, the first sector of a disk containing boot code and the partition table.

Q5. **What is WinRE?**
Ans: Windows Recovery Environment, a set of repair tools used when Windows fails to start.

EXPERIMENT 8

Command-Line Disk Management Tools

AIM: To explore and practise command-line disk management tools.

REQUIREMENTS / APPARATUS

Windows Command Prompt (Run as Administrator), a spare USB drive or a virtual disk (safer), Linux terminal for part B.

PROCEDURE A: DISKPART (WINDOWS)

1. Open CMD as Administrator and type `diskpart`.
2. `list disk` shows all disks.
3. `select disk 1`. **Check the disk number carefully so that your main disk is not selected.**
4. `list partition` and `list volume`.
5. `create partition primary size=100` creates a 100 MB partition.
6. `format fs=ntfs label=DATA quick`, then `assign letter=Z`.
7. Optional: `extend size=50`, `shrink desired=30`, `delete partition`, `clean` (wipes the disk!), `exit`.

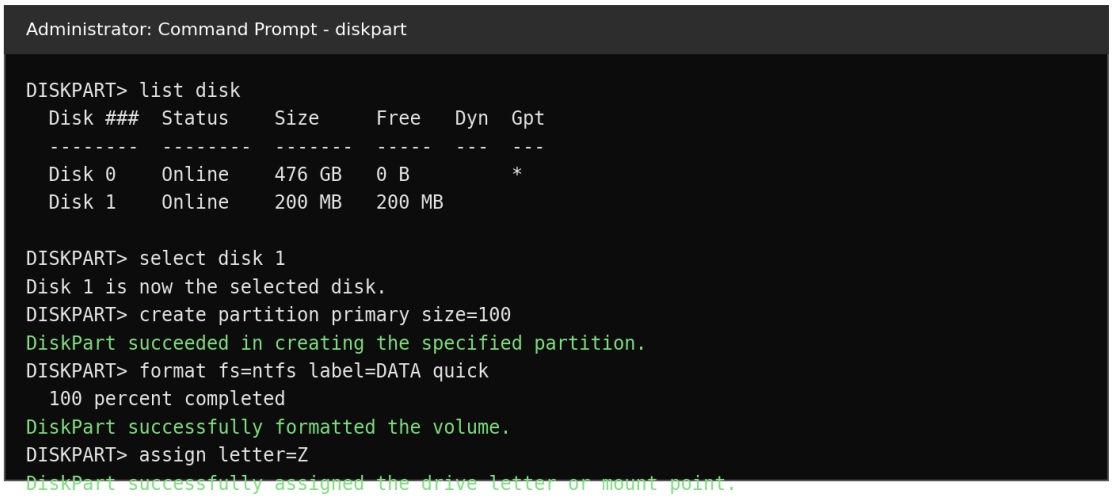


Fig 8.1: Creating and formatting a partition using DiskPart

Safe practice with a virtual disk:

```
create vdisk file="C:\test.vhd" maximum=200 type=expandable
attach vdisk
```

Other tools: `chkdsk D: /f` (check and fix), `defrag C: /O` (optimize), `fsutil volume diskfree C:`, `wmic diskdrive get model,size`.

PROCEDURE B: LINUX

```
lsblk                      # list block devices
sudo fdisk -l              # partition tables
df -h                      # disk usage by file system
du -sh folder              # size of a folder
sudo mkfs.ext4 /dev/sdb1   # format a partition
sudo mount /dev/sdb1 /mnt  # mount
sudo umount /mnt           # unmount
```

OBSERVATION

Record your own outputs.

Parameter	Value
Disk number selected	Disk 1
Disk size / free space	_____
New partition size	100 MB
File system	NTFS
Drive letter assigned	Z:

RESULT: Disk management tasks were performed successfully using command-line tools.

VIVA QUESTIONS

Q1. **What is DiskPart?**
Ans: A command-line utility for disk partitioning and management in Windows.

Q2. **What is a partition?**
Ans: A logical division of a physical disk.

Q3. **MBR vs GPT?**
Ans: MBR supports up to 2 TB and 4 primary partitions; GPT supports larger disks and up to 128 partitions.

Q4. **What is formatting?**
Ans: Preparing a partition with a file system so data can be stored.

Q5. **What does the clean command do?**
Ans: It erases all partition information of the selected disk, so use it very carefully.

EXPERIMENT 9

Process and Performance Monitoring

AIM: To monitor processes and system performance using Task Manager in Windows and process-related commands in Linux.

REQUIREMENTS / APPARATUS

Windows PC, Linux system (or WSL / virtual machine).

PROCEDURE A: WINDOWS (TASK MANAGER)

1. Press **Ctrl + Shift + Esc** to open Task Manager.
2. **Processes tab:** note the apps and background processes with CPU, Memory, Disk and Network usage.
3. **Performance tab:** view the CPU, Memory, Disk, Ethernet/Wi-Fi and GPU graphs.
4. **Startup tab:** disable unnecessary startup applications.
5. **Details / Services tab:** view the PID and the services.
6. Right-click an unresponsive application and choose **End task**.
7. CLI: `tasklist` and `taskkill /PID 1234 /F`.

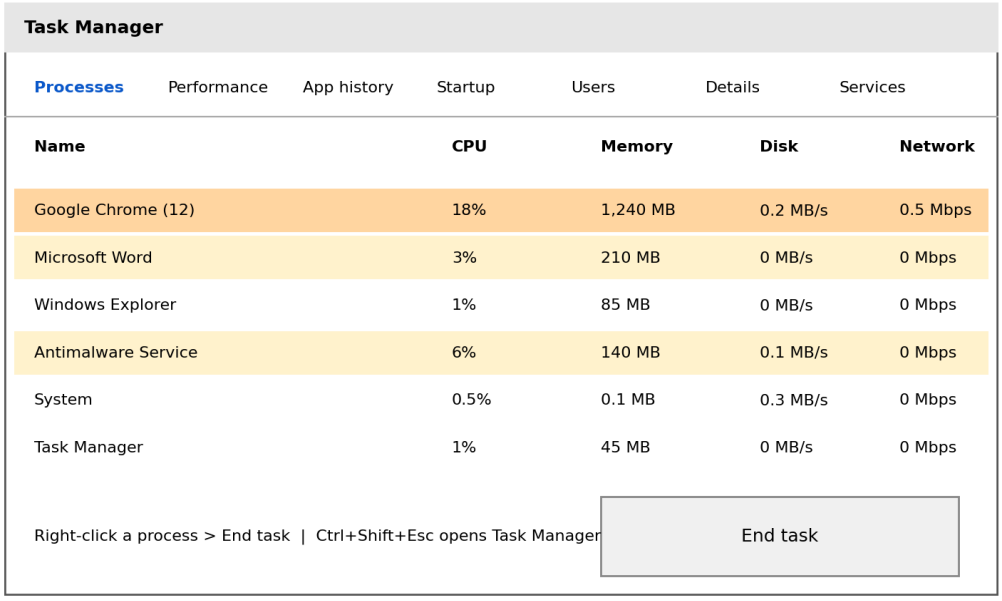


Fig 9.1: Task Manager - Processes tab (sample)

PROCEDURE B: LINUX

Command	Purpose
`ps aux`	List all running processes
`top` / `htop`	Live process monitor
`pgrep firefox`	Get the PID by name
`kill 1234` / `kill -9 1234`	Terminate / force-kill a process
`killall name`	Kill processes by name
`nice -n 10 cmd`, `renice 5 -p PID`	Set / change process priority
`jobs`, `bg`, `fg`	Manage background jobs
`free -h`, `uptime`, `vmstat`	Memory and load information

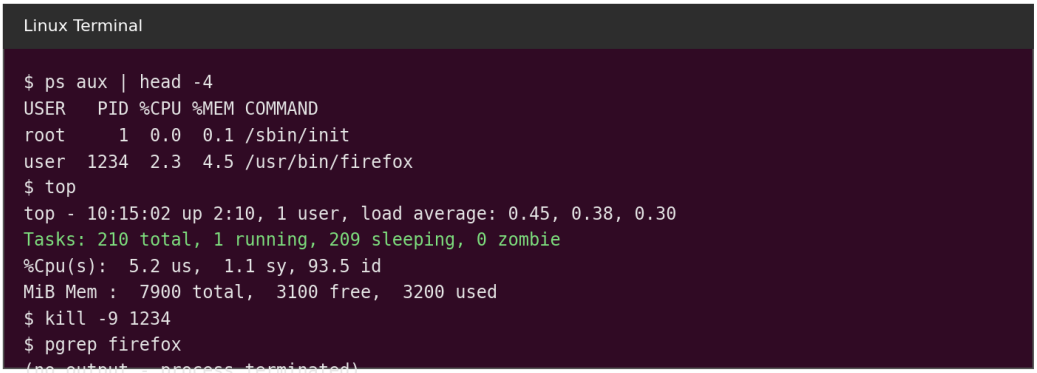


Fig 9.2: Linux process commands (ps, top, kill, pgrep)

OBSERVATION

Fill in the values from your own system.

Parameter	Value
CPU usage (idle)	_____ %
Memory used / total	_____ GB / _____ GB
Number of processes	_____
Highest CPU-consuming process	_____
PID of the process killed	_____

RESULT: Processes and system performance were monitored and managed using Task Manager and Linux commands.

VIVA QUESTIONS

Q1. **What is a process?**
Ans: A program in execution.

Q2. **What is a PID?**
Ans: The unique Process ID given to each process.

Q3. **Difference between kill and kill -9?**
Ans: kill sends SIGTERM (graceful stop); kill -9 sends SIGKILL (forced stop).

Q4. **What is the use of the Startup tab?**
Ans: To control the programs that launch automatically at boot.

Q5. **What is a zombie process?**
Ans: A finished process whose entry still remains in the process table.

EXPERIMENT 10

Network Configuration

AIM: To configure IP settings, set the DNS server, view network adapter information and enable/disable the network adapter.

REQUIREMENTS / APPARATUS

Windows PC with a network adapter (Ethernet/Wi-Fi), administrator rights, router / switch.

GUI route: run `ncpa.cpl`, right-click the adapter, Properties, select **Internet Protocol Version 4 (TCP/IPv4)**, Properties.

PROCEDURE

Task	GUI / Command
(a) Configure IP	Select *Use the following IP address* and enter IP 192.168.1.50, Subnet 255.255.255.0, Gateway 192.168.1.1. `netsh interface ip set address "Ethernet" static 192.168.1.50 255.255.255.0 192.168.1.1` Back to automatic: `netsh interface ip set address "Ethernet" dhcp`

(b) Set DNS	`netsh interface ip set dns "Ethernet" static 8.8.8.8`
	`netsh interface ip add dns "Ethernet" 8.8.4.4 index=2`
	Test: `nslookup google.com`

(c) View adapter info	`ipconfig /all`, `getmac`, PowerShell: `Get-NetAdapter`. Linux: `ip addr`, `nmcli device show`

(d) Enable / disable (Run as administrator)	`netsh interface set interface "Ethernet" disable`
	`netsh interface set interface "Ethernet" enable`
	GUI: right-click the adapter > Disable / Enable

Verification	`ping 192.168.1.1`, `ping 8.8.8.8`, `ping google.com`

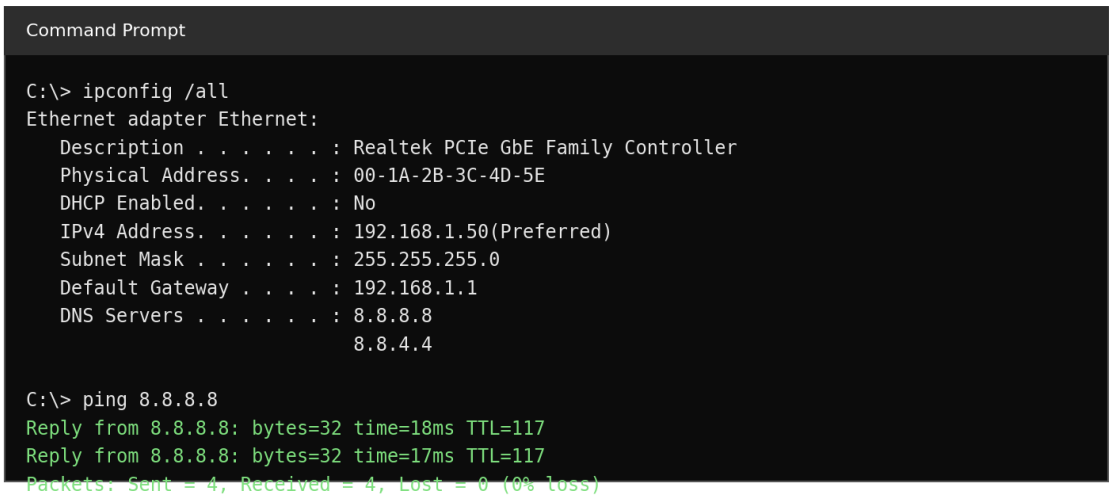


Fig 10.1: Adapter details after configuring a static IP and DNS, and ping test

OBSERVATION

Fill in from your own system.

Parameter	Value
Adapter name	Ethernet
MAC address	__ - __ - __ - __ - __ - __
IPv4 address (before / after)	_____ / 192.168.1.50

Parameter	Value
Subnet mask	255.255.255.0
Default gateway	192.168.1.1
DNS server	8.8.8.8, 8.8.4.4
Ping result	Replies received / Request timed out

After disabling the adapter its status shows **Disabled** and ping fails. After enabling it, connectivity returns.

Note: Use an IP that is free in your lab network (not used by another PC) to avoid an IP conflict. Restore DHCP after the experiment.

RESULT: IP settings and the DNS server were configured, adapter information was viewed, and the adapter was enabled and disabled.

VIVA QUESTIONS

Q1. **What is an IP address?**
Ans: A unique logical address that identifies a device on a network.

Q2. **What is DNS?**
Ans: A system that converts domain names into IP addresses.

Q3. **What is DHCP?**
Ans: A protocol that automatically assigns IP addresses to devices.

Q4. **What is a default gateway?**
Ans: The router that connects the local network to other networks.

Q5. **What is a MAC address?**
Ans: The 48-bit physical hardware address of the network interface card.

EXPERIMENT 11

Installing and Using a VPN

AIM: To download and install a third-party VPN application and establish an internet connection through the VPN.

REQUIREMENTS / APPARATUS

Internet connection, administrator rights, an official VPN application (e.g. ProtonVPN Free or Windscribe Free).

PROCEDURE

1. Note your **public IP before the VPN** (search 'what is my IP' in a browser) and your location.
2. Download the VPN installer **only from the official website**.
3. Run the installer, accept the licence terms and click Install.
4. Create a free account and log in.
5. Choose a server (country) and click **Connect**.
6. When the status shows *Connected*, check the public IP again.
7. Browse a website and run a speed test to compare.
8. Click **Disconnect** and verify that the original IP address is back.

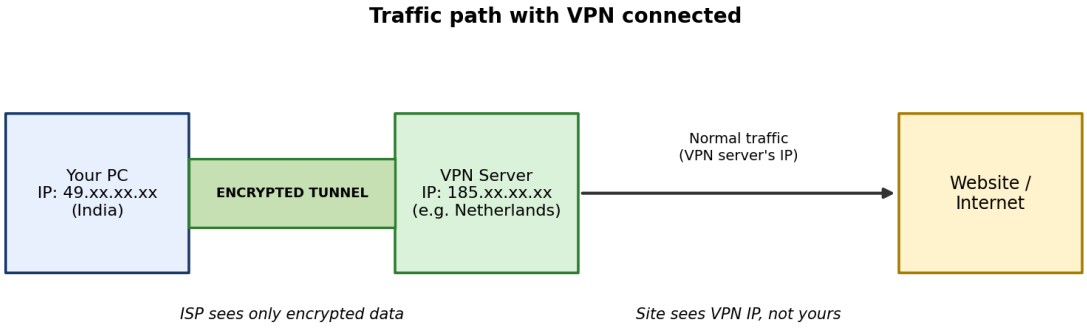


Fig 11.1: Traffic path when connected through a VPN

OBSERVATION

Parameter	Without VPN	With VPN
Public IP	_____	_____
Location	India	VPN server country
Speed / ping	_____	Usually slower
Data between PC and VPN server	Plain	Encrypted tunnel

The VPN creates an encrypted tunnel to its server. Websites see the VPN server's IP address instead of the user's real IP.

Note: Free VPNs may log data or show ads. College networks may block VPN ports; try another server or protocol if the connection fails.

RESULT: The VPN application was installed and an internet connection was established through the VPN.

VIVA QUESTIONS

Q1. **What is a VPN?**
Ans: Virtual Private Network, a secure encrypted connection over the internet.

Q2. **What are the uses of a VPN?**
Ans: Privacy, safe use of public Wi-Fi, remote office access and bypassing geo-restrictions.

Q3. **What is tunneling?**
Ans: Encapsulating packets inside another protocol for secure transport.

Q4. **Name some VPN protocols.**
Ans: OpenVPN, IKEv2/IPsec, WireGuard, L2TP.

Q5. **Why does the IP address change with a VPN?**
Ans: Because traffic exits to the internet through the VPN server.

EXPERIMENT 12

E-Waste Management (Visit / Case Study)

AIM: To analyse e-waste management processes through an on-site visit or a virtual case study.

REQUIREMENTS / APPARATUS

Visit permission letter or internet access for a virtual case study, notebook, camera / mobile (with permission).

PROCEDURE

1. Select a recycling facility (on-site visit) or a virtual case study, e.g. E-Parisaraa (Bengaluru), Attero Recycling or any CPCB/KSPCB-authorized recycler.
2. Study the process flow: collection, segregation, dismantling, material recovery (metals, plastics, glass) and safe disposal of hazardous parts (Fig 12.1).
3. Note the types of e-waste handled, the safety equipment used and the products recovered.
4. Study the applicable law: **E-Waste (Management) Rules, 2022** (in force from 1 April 2023), which includes Extended Producer Responsibility (EPR).
5. Prepare a short report with photos or screenshots.

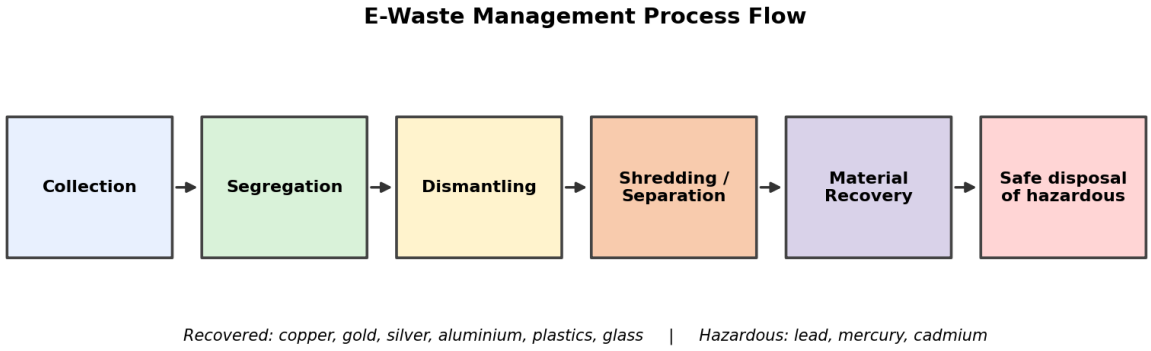


Fig 12.1: Typical e-waste recycling process flow

OBSERVATION

Item	Details
Facility / case study	_____
E-waste types	PCs, mobiles, batteries, TVs, cables, printers
Hazardous materials	Lead, mercury, cadmium, brominated flame retardants
Recovered materials	Copper, gold, silver, aluminium, plastic, glass
Safety measures	Gloves, masks, ventilation, controlled processing, trained staff
Legal framework	E-Waste (Management) Rules, 2022; EPR

RESULT: The e-waste management process was studied and the importance of recycling and safe disposal was understood.

VIVA QUESTIONS

Q1. What is e-waste?
Ans: Discarded electrical and electronic equipment.

Q2.	**Why is e-waste harmful?** Ans: It contains toxic metals that contaminate soil and water.
Q3.	**What are the 3 R's?** Ans: Reduce, Reuse, Recycle.
Q4.	**What is EPR?** Ans: Extended Producer Responsibility, which makes manufacturers responsible for the end-of-life of their products.
Q5.	**Why should e-waste not be burned?** Ans: It releases toxic fumes and dioxins.

EXPERIMENT 13

Computer Laboratory Setup Plan

AIM: To develop a comprehensive computer laboratory setup plan covering hardware procurement, installation, electrical safety measures and cybersecurity protocols.

REQUIREMENTS / APPARATUS

Sample requirement: a lab for 30 students + 1 instructor. Graph paper / drawing tool for the layout.

1. REQUIREMENT ANALYSIS AND LAYOUT

Room size (e.g. 30 x 20 ft), seating in rows with 3-4 ft aisles, the instructor's desk facing the students, proper ventilation/AC, lighting and an emergency exit.

Computer Lab Layout - 30 Students + 1 Instructor (approx. 30 ft x 20 ft)

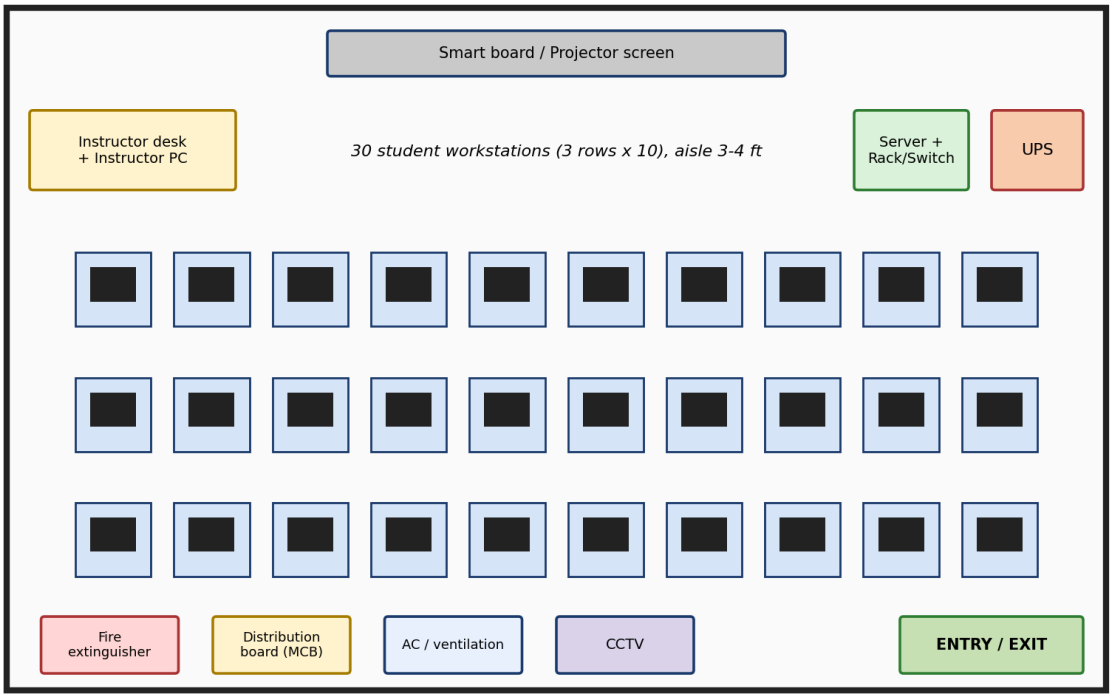


Fig 13.1: Proposed computer lab layout

2. HARDWARE PROCUREMENT

Item	Qty	Specification (example)
Desktop PCs	30 + 1	i5 / 8-16 GB RAM / 512 GB SSD / 21 inch monitor
Server	1	Lab management, file sharing and backup
Network switch	2	24-port managed
Router / firewall	1	With firewall features
UPS	1	Centralized online UPS, 10 kVA
Projector / smart board	1	For instructor demonstrations
Printer	1	Network laser printer
Furniture, cables, rack	-	Cat6 cable, patch panel, 12U rack, tables, chairs

Procurement steps: needs assessment, budget approval, quotations / tender (e.g. GeM portal), comparison, purchase order, delivery inspection, warranty / AMC record.

3. INSTALLATION

Assemble the PCs, install the OS from a master image (Ghost / Clonezilla), install software (compilers, Office, browsers), do structured Cat6 cabling and label every system, assign static IP / DHCP and test connectivity.

4. ELECTRICAL SAFETY MEASURES

- Proper **earthing** (earth resistance below 1 ohm is ideal for IT labs).
- MCB / ELCB (RCCB) at the distribution board and surge protectors.
- Separate power and data cable trays to avoid interference.
- UPS with backup; no overloaded sockets or loose wires.
- CO_2 fire extinguisher, smoke detector and first-aid kit.
- Anti-static flooring and temperature control (about 22-25 °C).
- Display safety instructions and carry out periodic inspections.

5. CYBERSECURITY PROTOCOLS

- Firewall, antivirus and regular OS / patch updates.
- Individual user accounts, strong password policy; admin rights only for staff.
- Web content filtering and USB port control.
- Network segmentation (student / staff VLAN).
- Regular data backups (3-2-1 rule) and login / activity logs.
- CCTV, visitor register and cyber-awareness sessions for students.

OBSERVATION

Phase	Duration (approx.)
Procurement	3-4 weeks
Installation and cabling	1 week
Testing and handover	2-3 days

The plan includes the layout diagram, bill of materials, timeline, and the safety and security checklists.

RESULT: A complete computer laboratory setup plan was prepared covering hardware, installation, electrical safety and cybersecurity.

VIVA QUESTIONS

Q1. **What is a UPS?**
Ans: Uninterruptible Power Supply, which gives backup power during outages.

Q2. **Why is earthing needed?**
Ans: To protect people and equipment from leakage current and electric shock.

Q3. **What is a firewall?**
Ans: A system that filters network traffic according to security rules.

Q4. **What is the 3-2-1 backup rule?**
Ans: Keep 3 copies of data on 2 different media with 1 copy offsite.

Q5. **What is AMC?**
Ans: Annual Maintenance Contract.